

First measurement of Rényi entanglement entropy in G_2 lattice gauge theory and comparison with $SU(3)$

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We present the first lattice measurement of the second Rényi entanglement entropy S_2 in pure G_2 Yang–Mills theory in four dimensions, and compare it with $SU(3)$ at matched string tension $\sqrt{\sigma}a \approx 0.22$. Using the replica trick with α -integration on lattices $L^3 \times 2L$ for $L = 4, 6, 8, 10, 12$, we find that S_2 obeys an area law $S_2 \propto L^{2.99}$ for both gauge groups. The ratio $S_2^{G_2}/S_2^{SU(3)}$ is 1.292 ± 0.001 , constant across all volumes, significantly below the naive gluon-counting prediction $\dim(\text{adj}_{G_2})/\dim(\text{adj}_{SU(3)}) = 14/8 = 1.75$. The 26% deficit is consistent with the Donnelly–Wall decomposition of gauge theory entanglement entropy: since the center $Z(G_2) = \{1\}$ is trivial, the classical (superselection) contribution to S_2 vanishes, whereas $SU(3)$ with $Z(3) \neq \{1\}$ retains this term. This constitutes the first *ab initio* lattice evidence for center-dependent entanglement entropy in non-abelian gauge theories.

I. INTRODUCTION

Entanglement entropy (EE) in gauge theories has attracted intense interest as a probe of confinement, topological order, and holographic duality [1–3]. A central open question is how EE depends on the gauge group G , and in particular on its center $Z(G)$.

The Donnelly–Wall decomposition [1] splits gauge theory EE into contributions from classical superselection sectors (determined by the center flux through the entangling surface), non-abelian edge modes, and bulk correlations. For a gauge group with trivial center $Z(G) = \{1\}$, the classical term vanishes identically. The exceptional Lie group $G_2 = \text{Aut}(\mathbb{O})$, the automorphism group of the octonions, provides a unique laboratory: it is the smallest simple Lie group with trivial center, it confines despite the absence of center symmetry [4, 5], and its adjoint representation has dimension 14.

In this work we present the first lattice measurement of the Rényi-2 entanglement entropy in pure G_2 Yang–Mills theory and compare it with $SU(3)$ at matched physical scale, providing the first quantitative test of the center-dependent EE decomposition.

II. METHOD

A. Lattice setup

We employ the standard Wilson plaquette action

$$S = \frac{\beta}{d_R} \sum_{\square} \text{Re Tr} (1 - U_{\square}), \quad (1)$$

where d_R is the dimension of the fundamental representation ($d_R = 7$ for G_2 , $d_R = 3$ for $SU(3)$), and

$\beta = d_R \cdot C_2(\text{fund})/g^2$. For G_2 , $\beta = 7/g^2$; for $SU(3)$, $\beta = 6/g^2$.

G_2 link variables are 7×7 real orthogonal matrices in the fundamental representation. The 14 generators of the Lie algebra $\mathfrak{g}_2 \subset \mathfrak{so}(7)$ are constructed as the null space of the constraint matrix enforcing preservation of the Bryant associative 3-form $\varphi = e^{123} + e^{145} + e^{167} + e^{246} - e^{257} - e^{347} - e^{356}$, yielding exactly 14 independent antisymmetric 7×7 matrices via singular value decomposition of the 35×21 linear system.

Updates are performed via single-link Metropolis with proposals $U' = e^{\varepsilon X} U$, where $X = \sum_{a=1}^{14} \alpha_a T_a$ with $\alpha_a \sim \mathcal{N}(0, \varepsilon^2)$ and $\varepsilon = 0.15$, achieving acceptance rates $\sim 40\%$ in the scaling window.

B. Entanglement entropy via replica trick

We measure the Rényi-2 entropy $S_2 = -\log \text{Tr}(\rho_A^2)$ using the α -integration method of Buividovich and Polikarpov [2]:

$$S_2 = \int_0^1 d\alpha \left\langle \frac{\partial S}{\partial \alpha} \right\rangle_{\alpha}, \quad (2)$$

where α parametrizes the interpolation between normal ($\alpha = 0$) and swapped ($\alpha = 1$) boundary conditions on the entangling surface ∂A , located at $x_0 = L/2$.

The lattice geometry is $L^3 \times 2L$ with periodic boundary conditions. The integral is evaluated by trapezoidal rule over 11 equally spaced α -points, each thermalized with 500 sweeps and measured over 200 configurations separated by 5 sweeps.

C. Matching physical scale

To compare G_2 and $SU(3)$ at the same physical scale, we match the string tension. From Bruno *et al.* [6], G_2 at $\beta = 10.0$ gives $\sigma a^2 = 0.047(1)$. For $SU(3)$, the same

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TABLE I. Rényi-2 entanglement entropy per unit area S_2/Area for G_2 ($\beta = 10.0$) and $SU(3)$ ($\beta = 6.06$) at matched string tension $\sigma a^2 \approx 0.047$. $\text{Area} = L^2 \times 2L$.

L	S_2/A [G_2]	S_2/A [$SU(3)$]	Ratio	14/8
4	18.303(12)	14.169(10)	1.292	1.750
6	18.113(7)	14.026(7)	1.291	1.750
8	18.123(6)	14.033(6)	1.291	1.750
10	18.080(10)	14.002(8)	1.291	1.750
12	18.087(10)	13.985(8)	1.293	1.750

$\sigma a^2 \approx 0.047$ corresponds to $\beta \approx 6.06$ (Necco–Sommer interpolation).

III. VALIDATION

A. Plaquette

The average plaquette $\langle P \rangle = (1/d_R) \text{Re Tr}(U_\square)$ at $\beta_{G_2} = 10.0$ converges to $\langle P \rangle = 0.561(1)$ for $L = 8$, consistent with the scaling window $\beta > 9.45$ identified in Ref. [6]. The β -scan over $\beta \in [9.6, 10.4]$ shows monotone increase of $\langle P \rangle$ with β and convergence between $L = 6$ and $L = 8$, confirming controlled finite-size effects.

For $SU(3)$ at $\beta = 6.06$, we obtain $\langle P \rangle = 0.619(1)$, consistent with literature values.

B. Area law

A power-law fit $S_2 \propto L^\gamma$ yields $\gamma = 2.99(1)$ for G_2 and $\gamma = 3.00(1)$ for $SU(3)$, confirming the expected area law $S_2 \propto \text{Area}(\partial A) = L^2 \times 2L = 2L^3$ in four dimensions.

IV. RESULTS

Table I presents the central result. The ratio $R = S_2^{G_2}(L)/S_2^{SU(3)}(L)$ is remarkably constant:

$$R = 1.292 \pm 0.001 \quad (\text{all } L). \quad (3)$$

This is 26% below the naive gluon-counting prediction $R_{\text{dim}} = \dim(\text{adj}_{G_2})/\dim(\text{adj}_{SU(3)}) = 14/8 = 1.75$.

Both datasets are well described by the two-parameter fit

$$\frac{S_2}{\text{Area}} = a_\infty + \frac{c}{L^2}, \quad (4)$$

with $a_\infty^{G_2} = 18.04(2)$, $c^{G_2} = 3.9(6)$ and $a_\infty^{SU(3)} = 13.97(2)$, $c^{SU(3)} = 3.1(5)$. The ratio of asymptotic coefficients $a_\infty^{G_2}/a_\infty^{SU(3)} = 1.292$ coincides with the L -independent ratio, confirming that the deficit is not a finite-size artifact.

V. DONNELLY–WALL ANALYSIS

A. Hypothesis testing

We test four hypotheses for the scaling of S_2/A with gauge group:

H₀: Gluon counting. If each gluon contributes equally, $R = \dim(\text{adj}_{G_2})/\dim(\text{adj}_{SU(3)}) = 1.75$. *Rejected*: measured $R = 1.29$, deficit 26%.

H₁: Center contribution (Donnelly–Wall). In the decomposition of Ref. [1], $Z(G_2) = \{1\}$ eliminates the classical superselection contribution, while $SU(3)$ retains $S_{\text{classical}} \propto \log|Z(3)|$. *Consistent*: the measured deficit is in the direction predicted by center suppression.

H₂: Universal a_∞ . If the UV-divergent coefficient is group-independent, $a_\infty^{G_2} = a_\infty^{SU(3)}$. *Rejected*: they differ by 29%.

H₃: Dimensional decomposition. The ratio of $1/L^2$ coefficients $c^{G_2}/c^{SU(3)} = 1.26 \approx R$, suggesting that both the leading and sub-leading terms scale with the same effective dimension.

B. Effective dimension

The constant ratio $R = 1.292$ implies an effective number of entangling gluon degrees of freedom:

$$d_{\text{eff}}(G_2) = R \times d_{\text{adj}}(SU(3)) = 1.292 \times 8 \approx 10.3. \quad (5)$$

Out of 14 adjoint gluons in G_2 , approximately 10.3 contribute to the entanglement across the boundary. The 3.7 “missing” gluons are consistent with partial screening of the $\mathbf{3} \oplus \bar{\mathbf{3}}$ sector under the maximal $SU(3) \subset G_2$ decomposition $\mathbf{14} = \mathbf{8} \oplus \mathbf{3} \oplus \bar{\mathbf{3}}$: the six gluons transforming in the (anti-)fundamental of $SU(3)$ can form color singlets and are partially screened at the entangling surface.

VI. DISCUSSION

A. Confinement without center

G_2 confines despite its trivial center [4], with a first-order deconfinement transition [5, 6] driven by mechanisms other than center vortex condensation. Our measurement demonstrates that the entanglement structure of the confining vacuum is correspondingly modified: the area law coefficient is reduced relative to the dimensional expectation, consistent with the absence of center-vortex-mediated long-range correlations.

B. Dark sector phenomenology

G_2 has been proposed as a dark gauge group [7], with dark matter candidates arising from G_2 glueball

states. Our measurement of S_2/A provides the first non-perturbative input for the information-theoretic properties of such a dark sector: the reduced entanglement relative to $SU(3)$ suggests weaker correlations in the dark vacuum, with potential implications for dark matter self-interactions.

C. M-theory context

G_2 holonomy manifolds play a central role in M-theory compactifications to four-dimensional $\mathcal{N} = 1$ theories [8]. While the connection between G_2 as a gauge group and G_2 as holonomy is not direct, our lattice data provides the first empirical constraint on the entanglement structure of G_2 gauge dynamics.

D. Limitations

This measurement has several limitations: (i) single lattice spacing (β fixed), preventing continuum extrapolation; (ii) Rényi-2 entropy rather than von Neumann; (iii) half-lattice entangling surface only; (iv) Metropolis algorithm (not HMC), with potential autocorrelation issues at $L \geq 12$. A multi- β study and HMC implemen-

tation are planned for future work.

VII. CONCLUSION

We have performed the first lattice measurement of entanglement entropy in G_2 gauge theory and compared it with $SU(3)$ at matched physical scale. The ratio $S_2^{G_2}/S_2^{SU(3)} = 1.292 \pm 0.001$ is constant across lattice sizes $L = 4$ –12 and 26% below the gluon-counting prediction of 1.75. This deficit provides the first *ab initio* lattice evidence for the center-dependent contribution to gauge theory entanglement entropy predicted by the Donnelly–Wall decomposition. Our result establishes G_2 gauge theory as a unique laboratory for studying the interplay between gauge symmetry, center structure, and quantum entanglement.

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